



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (E&E) (Credit System) Degree Examinations
Supplementary Examinations - December 2025

21EE502 / 20EE502 - SIGNAL ANALYSIS AND PROCESSING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I

1. a) What is meant by Nyquist sampling rate? Explain with an example.
 b) Determine the odd and even components of the signal shown in Figure Q1.b

Marks	BT	CO	PO
6	L*1	1	1,2,10,12

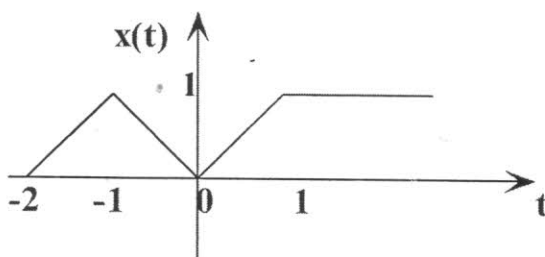


Fig. Q1.b

- c) Determine the convolution sum of the sequences:
 $x_1[n] = \{1 \ 2 \ 1 \ -1\}$ and $x_2[n] = \{1 \ 2 \ 3 \ 1\}$
2. a) Explain the basic operations of time shifting and time scaling on signals.
 b) Determine whether the given signal is periodic or not. If periodic what is the fundamental period?
 i) $x(t) = \cos\left(\frac{\pi}{3}\right)t + \sin\left(\frac{\pi}{4}\right)t$
 ii) $x[n] = \cos\left(\frac{1}{5}\pi n\right) \sin\left(\frac{1}{3}\pi n\right)$
- c) Find the solution of the difference equation given below by using the prescribed initial conditions:
 $Y[n] - 0.6y[n-1] = x[n] = (0.4)^n, n \geq 0, y(-1) = 10$
3. a) Explain the convolution principle in discrete time.
 b) Determine whether the following systems are memoryless, stable and causal: i) $y(t) = 2x(t) + 10$ ii) $y[n] = nx[n]$
 c) Perform the following operations on the signal shown in Figure Q3.c

6	3	1	1,2,10,12
8	3	2	1,2,10,12
6	2	1	1,2,10,12
6	3	1	1,2,10,12
8	4	2	1,2,10,12
6	2	2	1,2,10,12
6	3	2	1,2,10,12

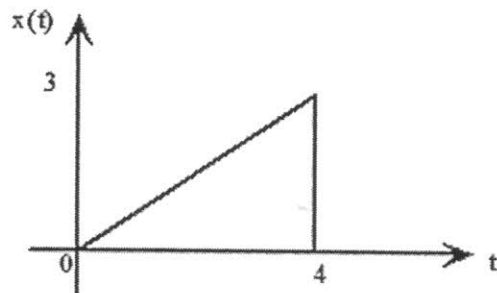


Fig. Q3.c

- (i) $x(3t+2)$ (ii) $x(2(t+2))$ (iii) $x(-2t-1)$ (iv) $x(-t)$

8	4	2	1,2,10,12
---	---	---	-----------

P.T.O.

Unit – II

4. a) Find DTFS Coefficients for the waveform shown in Figure Q4.a

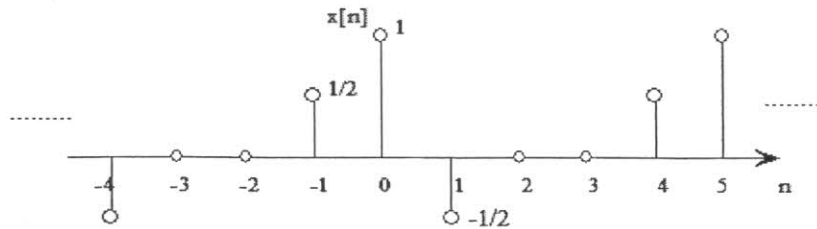


Fig. Q4.a

- b) Find the 4 point DFT of a sequence $x(n) = \{0 \ 1 \ 2 \ 3\}$ and hence sketch its magnitude and phase plots.
- c) State and prove the following properties of DFT.
- Linearity
 - Multiplication of two DFTs
5. a) Compare Linear Convolution and Circular Convolution.
- b) Consider a causal and stable LTI system S whose input $x[n]$ and output $y[n]$ are related through the second order difference equation using DTFT and inverse DTFT method
- $$y[n] - \frac{1}{6}y[n-1] - \frac{1}{6}y[n-2] = x[n]$$
- Determine the frequency response $H(e^{j\omega})$ for the system S
 - Determine the impulse response $h[n]$ for the system S
- c) With the help of signal flow graph, explain the radix-2 decimation in frequency FFT algorithm.
6. a) Consider the length 6 sequence defined for $0 \leq n < 6$ $x[n] = \{1, -2, 3, 0, -1, 1\}$ with a 8 point DFT $X(k)$. Evaluate the following functions of $X(k)$ without computing DFT:
- $X(0)$
 - $X(3)$
 - $\sum_{k=0}^5 |X(k)|^2$
- b) Let $x(n)$ be 4 point sequence of $x(n) = \{1 \ 1 \ 0 \ 0\}$. Compute DFT of the sequence using DIT FFT algorithm.
- c) Explain the principle of circular convolution of two sequences. Hence find circular convolution of two sequences $X_1[n] = \{2 \ 1 \ 2 \ 1\}$ and $x_2[n] = \{1 \ 2 \ 3 \ 4\}$

Unit – III

7. a) Obtain the Direct form-I, Direct form-II form realizations of the LTI system governed by the equation
- $$y[n] + \frac{1}{4}y[n-1] - \frac{1}{8}y[n-2] = x[n] + x[n-1]$$
- b) Explain the design principles of low pass digital IIR filter using Butterworth approximation.
8. a) Realize the following linear phase systems
- $H(z) = \frac{1}{3} + \frac{1}{5}z^{-1} + \frac{2}{3}z^{-2} + \frac{1}{5}z^{-3} + \frac{1}{3}z^{-4}$
 - $H(z) = \frac{1}{2} + \frac{1}{4}z^{-1} + \frac{1}{4}z^{-2} + \frac{1}{2}z^{-3}$
- b) Design an FIR filter with
- $$H_d(e^{j\omega}) = \begin{cases} e^{-2j\omega} & -\frac{\pi}{4} \leq \omega \leq \frac{\pi}{4} \\ 0 & \frac{\pi}{4} \leq \omega \leq \pi \end{cases}$$
- Determine the filter coefficients of $h_d[n]$. Also determine the response $H(\omega)$

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
